

MIORA

For the ones on your mind.

MIORA: A People-Centered Memory Journal

Abstract

Meaningful relationships are not defined solely by how frequently people communicate. Individuals may experience long gaps in communication with friends, family members, or other important people while still thinking about them, remembering shared experiences, recalling conversations, or associating them with particular places, songs, and emotions. However, these moments of remembrance are often temporary and remain unrecorded or scattered across notes, photographs, messages, and other platforms.

MIORA is proposed as a people-centered, thought-first memory journal that provides users with a private space to capture moments when important people come to mind. Users can record a simple thought event and optionally associate it with memories, conversations, places, feelings, and other meaningful context. Over time, these entries create a personal timeline representing the memories and emotional connections associated with each person.

The proposed system is based on the idea that a lack of regular communication does not necessarily indicate a lack of care, remembrance, or emotional connection. By organizing memories around people rather than only dates or interactions, MIORA enables users to preserve meaningful moments and reflect on the people who continue to remain in their thoughts.

MIORA is designed primarily as a private personal memory space. In future versions, users may optionally choose to share selected memories or collections with important people, allowing them to express that they were remembered even during periods of distance or silence.

1. Introduction

Human relationships are not defined only by conversations, messages, calls, or meetings. A person can remain important in someone's life even when there are long periods without communication. Friends may lose regular contact because of busy schedules, distance, work, studies, or changes in life, while still remembering each other frequently.

A familiar place, an old conversation, a song, a photograph, or an unexpected moment can suddenly bring someone to mind. These moments may create feelings of happiness, nostalgia, gratitude, sadness, or a desire to reconnect. However, such moments are usually temporary. They may remain only in an individual's memory and gradually fade with time.

Existing communication platforms primarily record direct interactions between people. They can show when two people last communicated but cannot represent how often someone was remembered during a period of silence. Similarly, traditional journals allow users to record personal experiences but generally organize entries around dates rather than around the people connected to those experiences.

This creates a gap between communication history and emotional remembrance. A long period without messages does not necessarily mean that a person has been forgotten or has become less important.

MIORA is proposed to address this gap through a people-centered and thought-first approach to personal memory journaling. The application allows users to maintain a private collection of important people and capture moments when those individuals come to mind. A user can simply record that they thought about someone and optionally add information about the memory, conversation, place, feeling, or event associated with that moment.

Over time, these entries create a structured memory history for each person. Users can revisit meaningful experiences and reflect on the people, memories, and emotions that continue to hold significance in their lives.

MIORA is not intended to measure relationships through communication frequency or encourage comparison between people. Instead, it provides a private space for recognizing and preserving the fact that meaningful relationships can continue to exist in thoughts and memories, even during periods of distance or silence.

The central idea behind MIORA can therefore be expressed as:

Not talking to someone does not always mean that they are forgotten.

2. Problem Statement

People often continue to think about important individuals even when regular communication is absent. A person may go weeks or months without speaking to a friend or family member while still remembering them through past conversations, meaningful places, shared experiences, songs, photographs, or everyday events.

However, existing digital platforms primarily focus on recording direct interactions or organizing personal experiences by date. Communication applications can indicate when people last spoke but cannot capture moments when someone was remembered without communication. Similarly, traditional journals and note-taking applications do not provide a dedicated person-centered approach for organizing these moments of remembrance.

As a result, meaningful thoughts, memories, and emotional connections associated with important people may remain scattered, unrecorded, or gradually fade over time. Additionally, individuals may have no meaningful way to look back and recognize how often someone remained in their thoughts during periods of distance or silence.

Therefore, there is a need for a private, people-centered system that enables individuals to capture moments when important people come to mind and preserve the memories, conversations, places, feelings, and experiences associated with those moments.

MIORA addresses this need through a thought-first memory journaling approach, allowing users to record:

Person → Thought → Memory and Context

The system aims to preserve meaningful personal connections while recognizing that:

A gap in communication does not necessarily represent a gap in care or remembrance.

Core Problem Question

How can individuals privately capture and preserve moments when important people come to mind, along with the memories and emotions associated with those moments, regardless of how frequently they communicate?

3. Feasibility Study

3.1 Technical Feasibility

MIORA is technically feasible to develop using modern web or mobile application technologies. The core system primarily involves user authentication, management of people, recording thought events, storing memories, associating feelings and places, and presenting information through timelines and personal insights.

The system can be developed using a frontend application, backend API, and database architecture. The initial version does not require advanced artificial intelligence or complex infrastructure. AI-based features can be introduced later if required for automatically identifying relationships between memories, people, places, and emotions.

Therefore, the technical feasibility of the proposed system is considered high.

3.2 Economic Feasibility

The initial development cost of MIORA is expected to be low. Open-source technologies and free-tier cloud services can be used during the development and testing stages. Since the initial application primarily stores personal text-based memories and structured information, the infrastructure requirements are also relatively small.

The major resources required are development time, database storage, authentication services, and hosting.

Therefore, the proposed system is considered economically feasible, particularly as a student project and early-stage product.

3.3 Operational Feasibility

MIORA is designed around a simple and natural user interaction. When a person comes to the user's mind, the user should be able to quickly record the moment without completing a lengthy journal entry.

The core interaction can be as simple as:

Select a Person → "I Thought About Them" → Save

Additional information such as a memory, place, conversation, or feeling can be added optionally.

This low-friction approach makes the system easier to integrate into everyday life. However, long-term usage depends on ensuring that the application remains simple and does not make recording memories feel like a task or obligation.

Therefore, MIORA is considered operationally feasible, provided the user experience remains lightweight and intuitive.

3.4 Privacy and Legal Feasibility

MIORA may store personal memories involving other individuals, including names, conversations, places, feelings, and personal experiences. Therefore, privacy must be considered a fundamental requirement.

The system should follow a privacy-first approach, including:

- Private user accounts
- Secure authentication
- Personal data accessible only to the account owner
- No public profiles by default
- No public sharing without user permission
- Ability to modify or delete stored information

The application should function primarily as a user's personal memory journal and should not imply that individuals mentioned in the journal are users of the system.

Therefore, the project is considered legally and operationally feasible with appropriate privacy and data protection practices.

3.5 Schedule Feasibility

MIORA can be developed in phases.

The initial version can focus on the core functionality:

- Managing people
- Recording thought events
- Adding memories
- Recording feelings and places
- Viewing person-centered timelines

Advanced features such as intelligent memory analysis, relationship insights, and optional sharing can be developed in later versions.

A phased development approach makes the project achievable within a reasonable development schedule.

Feasibility Conclusion

Based on the technical, economic, operational, privacy, and schedule considerations, MIORA is considered a feasible project. The primary challenge is not the technical implementation but ensuring that users find sufficient long-term value in recording and revisiting these personal moments.

4. Market and User Feasibility

4.1 User Need

People often think about important individuals even when they are not in regular communication with them. Friends, family members, and other meaningful people may come to mind through memories, conversations, places, songs, photographs, or unexpected moments.

A person may go several months without speaking to someone while still remembering them many times during that period. However, these moments are usually not recorded.

MIORA addresses this experience by providing a person-centered space for capturing such moments.

4.2 Target Users

The primary target users of MIORA are individuals who:

- Have meaningful relationships with a small number of important people
- May experience long gaps between conversations
- Frequently remember people from their lives
- Associate people with memories, places, conversations, or emotions
- Want to preserve meaningful personal experiences

MIORA may particularly appeal to people who are not highly communicative but still maintain strong emotional connections and memories with important individuals.

4.3 Current Alternatives

Currently, users may preserve personal memories using:

- Notes applications
- Traditional journals
- Photo galleries
- Saved messages
- Social media

- Personal CRM applications

However, these alternatives do not directly focus on the question:

"Who came to my mind today?"

Traditional journals are generally organized around dates, while communication applications focus on direct interactions. Personal CRM applications are often designed to help users remember information about people and maintain relationships.

MIORA introduces a different approach by organizing memories around people and beginning with a thought event.

4.4 MIORA's Core Differentiation

MIORA follows a thought-first approach.

The core interaction begins with:

"I thought about this person."

The user can then optionally record:

- A memory
- A conversation
- A place
- A feeling
- A trigger such as a song or photograph

This creates the following model:

Person → Thought → Memory and Context

Instead of focusing only on when users interact with someone, MIORA captures when someone naturally remains in the user's thoughts.

4.5 User Motivation and Long-Term Value

The value of MIORA is not limited to recording memories for personal reflection.

Over time, users may build a meaningful record showing that certain people remained important to them even during periods without communication.

For example, a user may someday be able to say:

"We may not have talked often, but I still remembered you."

The accumulated thoughts and memories can provide a personal representation of care, remembrance, and emotional connection.

In the future, MIORA may allow users to optionally share selected memories or collections with someone important. However, the application will remain private by default.

4.6 Market and User Feasibility Conclusion

The identified user need is emotionally relatable and reflects a common experience: remembering important people despite long periods without communication.

MIORA's primary differentiation is its thought-first, person-centered approach to memory journaling. While alternative applications exist for journaling, memory preservation, and relationship management, MIORA focuses specifically on capturing the moments when people naturally come to mind.

The main adoption challenge is ensuring that recording these moments remains quick and effortless. Therefore, the core experience must prioritize low-friction interaction.

Overall, the market and user feasibility of MIORA is considered promising, with the thought-first concept providing a clear direction for the product.

5. Objectives of the Proposed System

5.1 Main Objective

The main objective of MIORA is to develop a private, people-centered memory journal that enables users to capture, preserve, and reflect on moments when important people come to mind.

5.2 Specific Objectives

The specific objectives of the proposed system are:

1. To enable users to maintain a personal collection of people who are meaningful to them.
2. To provide a quick and simple method for recording moments when an important person comes to mind.
3. To track and preserve the frequency of remembrance associated with each person over time.
4. To allow users to associate moments of remembrance with memories, conversations, places, feelings, and other meaningful context.
5. To create person-centered memory timelines that allow users to revisit thoughts and experiences associated with important people.
6. To support personal reflection on meaningful relationships beyond the frequency of direct communication.
7. To ensure that personal memories and information remain private and under the user's control.

6. Scope of the Proposed System

The scope of MIORA defines the features included in the initial version of the system, the features intentionally excluded from the current implementation, and possible enhancements for future development.

6.1 In Scope

The initial version of MIORA will focus on the core concept of person-centered, thought-first memory journaling.

People Management

Users will be able to create and maintain a personal collection of people who are meaningful to them. Basic information associated with each person can be managed by the user.

Thought and Remembrance Recording

Users will be able to quickly record moments when an important person comes to mind. Each recorded thought will contribute to the person's remembrance history and allow the system to maintain the frequency of remembrance over time.

Contextual Memory Capture

Users will be able to optionally associate a thought with meaningful context, including:

- Memories or personal notes
- Remembered conversations
- Places
- Feelings or emotions

Providing contextual information will remain optional to ensure that recording a thought remains quick and simple.

Person-Centered Memory Timeline

MIORA will provide a chronological timeline for each person, allowing users to revisit recorded thoughts, memories, and associated experiences.

Basic Personal Reflection

Users will be able to view basic information such as:

- Total number of times a person was remembered
- Recent moments of remembrance
- Person-centered memory history
- Associated memories, places, and feelings

Privacy and User Control

Personal information and memories will remain private and under the user's control. Users will be able to manage, edit, and delete their stored information.

6.2 Out of Scope

The initial version of MIORA will not include features that move the application away from its primary purpose as a private personal memory journal.

The following features are outside the scope of the initial version:

- Social media feeds
- Followers, likes, or comments
- Public profiles
- Messaging or communication functionality
- Automatic detection of thoughts
- Emotional or relationship ranking between people
- Photo storage and photo-based memories

MIORA is not intended to replace communication platforms or function as a social networking application.

6.3 Future Scope

The following features may be considered in future versions of MIORA:

- Photo-based memories
- AI-assisted identification of people, places, and emotions from written memories
- Advanced insights and memory visualizations
- Advanced connections between people, places, memories, and feelings

- Optional sharing of selected memories or memory collections with important people
- Personalized memory reminders and reflection features

These enhancements will be considered after validating and establishing the core thought-first memory journaling experience.

Scope Conclusion

The initial version of MIORA will remain focused on a simple core experience:

Person → Thought → Optional Memory Context → Timeline and Reflection

This focused scope will help ensure that the system remains simple, practical, and aligned with its primary purpose.

7. User Personas and User Journey

7.1 User Personas

MIORA is designed around user behavior and personal needs rather than specific demographic groups. The primary users are individuals who value meaningful relationships and personal memories.

Persona 1: The Quiet Rememberer

The Quiet Rememberer is the primary target user of MIORA.

This user may care deeply about a small number of important people but may not communicate with them regularly. Long gaps between conversations may occur because of distance, work, studies, busy schedules, or personal communication habits. Despite these gaps, the user frequently remembers important people through places, conversations, songs, experiences, and everyday events.

The primary need of this user is to capture and preserve moments when someone comes to mind.

MIORA provides a simple way to record these moments and maintain a personal history of remembrance over time.

Persona 2: The Memory Keeper

The Memory Keeper values meaningful experiences and does not want important memories to be forgotten.

This user may currently store memories across notes, photographs, messages, and journals. However, these memories are often scattered and difficult to revisit in connection with a particular person.

The primary need of this user is to organize meaningful memories around the people associated with them.

MIORA provides a person-centered space where thoughts, memories, conversations, places, and feelings can be connected to an individual and revisited over time.

7.2 Primary User Journey

The primary user journey begins when an important person unexpectedly comes to the user's mind.

Step 1: Trigger

A person may come to mind because of:

- Visiting a familiar place
- Remembering a past conversation
- Hearing a song
- Seeing something familiar
- Recalling a shared experience
- Experiencing an unexpected moment of remembrance

Step 2: Open MIORA

The user opens MIORA and sees a people-centered interface containing the people who are important to them.

The primary question presented to the user is:

Who is on your mind?

Step 3: Select a Person

The user selects the person they are currently thinking about.

Step 4: Record the Thought

The user records the moment using a simple action such as:

I thought about them.

The system stores the event and updates the person's remembrance history.

Step 5: Add Optional Context

If desired, the user can add additional information associated with the thought, including:

- A personal memory or note
- A remembered conversation
- A place
- A feeling or emotion

Adding context is optional to ensure that recording a thought remains quick and effortless.

Step 6: Revisit and Reflect

At a later time, the user can visit the person's profile and view:

- Total remembrance count
- Recent moments of remembrance
- Associated memories
- Places
- Feelings
- A chronological memory timeline

This allows users to reflect on the meaningful people and experiences that remain present in their thoughts.

7.3 People-First Navigation

MIORA follows a people-first navigation approach.

The home experience begins with the user's collection of meaningful people rather than requiring the user to first create a journal entry.

The primary interaction flow is:

People → Select Person → Record Thought → Add Optional Context → Timeline

This approach reinforces MIORA's identity as a people-centered memory journal and provides users with a simple and natural way to capture moments of remembrance.

7.4 Future User Journey

In future versions, MIORA may allow users to optionally select and share meaningful memories or collections with an important person.

For example, a user may choose to share a collection representing moments when someone remained in their thoughts during a particular period.

This feature is outside the scope of the initial version and will remain completely optional. MIORA will continue to operate as a private personal memory space by default.

8. Functional Requirements

The functional requirements define the core capabilities and behaviors that the MIORA system must provide. The requirements are organized into functional modules.

8.1 User Authentication and Data Access

FR-01: User Registration and Authentication

The system shall allow users to create an account and securely access their personal MIORA space.

FR-02: User Data Isolation

The system shall ensure that users can access only their own people, thoughts, memories, and associated information.

8.2 People Management

FR-03: Add Person

The system shall allow users to add a person to their personal collection.

The required information shall be:

- Name

Optional information shall include:

- Nickname

A person in MIORA represents someone meaningful to the user and is not intended to function as a detailed contact profile.

FR-04: View People

The system shall display the user's active people through a people-first interface.

The primary home experience shall allow users to directly view and select people.

FR-05: Search People

The system shall allow users to search for people using their name or nickname.

FR-06: Edit Person

The system shall allow users to edit a person's name or nickname.

FR-07: Archive Person

The system shall allow users to archive a person instead of permanently deleting them.

When a person is archived:

- The person shall be removed from the active people list.
- Their associated thoughts, memories, and contextual information shall remain preserved.
- The person shall remain accessible through the archived people section.

FR-08: Restore Archived Person

The system shall allow users to restore an archived person to the active people collection.

8.3 Permanent Deletion Process

FR-09: Request Permanent Deletion

The system shall allow users to explicitly request the permanent deletion of an archived person and their associated information.

The system shall not permanently delete the information immediately.

FR-10: Deletion Pending Period

After a permanent deletion request, the system shall place the person and associated information into a deletion-pending state.

FR-11: Deletion Confirmation

The system shall provide a reminder after five days and request a final confirmation after fifteen days.

Permanent deletion shall occur only after explicit user confirmation.

If the user cancels the deletion request or does not provide explicit confirmation, the information shall remain archived.

8.4 Thought and Remembrance Recording

FR-12: Record Thought

The system shall allow users to record a moment when a selected person comes to mind.

The core action shall represent:

I thought about this person.

Each thought event shall be associated with the selected person and the user.

FR-13: Thought Timestamps

Each thought event shall store:

- The date and time when the thought occurred.
- The date and time when the thought was recorded in MIORA.

By default, the occurred date and time shall be set to the current date and time.

FR-14: Backdated Thought Recording

The system shall allow users to modify the occurred date and time when recording a thought.

This allows users to record a past moment of remembrance that was not immediately added to the system.

FR-15: Multiple Thoughts

The system shall allow users to record multiple genuine moments of remembrance for the same person, including multiple thoughts on the same day.

Each valid thought event shall contribute to the person's total remembrance count.

FR-16: Cooldown Period

The system shall support a configurable cooldown period for repeated thought recording.

The default cooldown period shall be five minutes.

FR-17: Accidental Repeated Tap Detection

When repeated thought actions occur within a very short period, the system shall detect possible accidental interactions and request confirmation from the user.

The user shall be able to indicate whether the repeated interaction represents a genuine additional remembrance.

If the interaction was accidental, only the valid original thought event shall be retained.

FR-18: Undo Thought

After recording a thought, the system shall provide an option to undo the recorded thought.

The system shall request confirmation before permanently removing the thought event.

8.5 Thought Context

All contextual information associated with a thought shall be optional.

FR-19: Add Memories or Notes

The system shall allow users to add one or more personal memories or notes associated with a thought.

FR-20: Add Remembered Conversations

The system shall allow users to record one or more conversations associated with a thought.

FR-21: Add Places

The system shall allow users to associate multiple places with a thought.

Places shall support free-text input.

FR-22: Add Feelings

The system shall allow users to associate multiple feelings or emotions with a thought.

Feelings shall support free-text input.

8.6 Person Profile and Memory Timeline

FR-23: Person Profile

The system shall provide a dedicated profile for each person.

The profile shall display relevant information, including:

- Person name
- Nickname, if available
- Total remembrance count
- Recent thought activity

FR-24: Person-Centered Timeline

The system shall provide a chronological timeline of thought events and associated contextual information for each person.

The timeline shall primarily use the date and time when the thought occurred.

FR-25: Review Associated Context

The system shall allow users to revisit memories, conversations, places, and feelings associated with a person's thought history.

8.7 Reflection and Basic Insights

FR-26: Remembrance Count

The system shall maintain and display the total number of valid thought events recorded for each person.

FR-27: Recent Activity

The system shall allow users to view recent moments of remembrance associated with a person.

The initial version shall focus on simple reflection rather than advanced analytical insights.

8.8 Data Management and Privacy

FR-28: Edit Contextual Information

The system shall allow users to edit contextual information associated with their thought events.

FR-29: Delete Thought Events

The system shall allow users to delete individual thought events and associated contextual information after confirmation.

FR-30: User Data Control

The system shall provide users with control over their personal information, including the ability to manage, archive, restore, edit, and delete their data.

Functional Requirements Summary

The core functional flow of MIORA is defined as:

People → Select Person → Record Thought → Add Optional Context → Timeline → Reflection

The system prioritizes quick thought capture while allowing users to gradually build a meaningful, person-centered history of memories and remembrance.

9. Non-Functional Requirements

Non-functional requirements define the quality attributes and operational characteristics of the MIORA system. These requirements describe how the system should perform and behave while providing its functional capabilities.

9.1 Performance

MIORA should provide a responsive user experience for common operations.

Activities such as viewing people, recording a thought, opening a person's profile, and viewing a timeline should normally complete within approximately two seconds under normal network conditions.

The system should provide immediate feedback when a thought event is successfully recorded.

9.2 Security

The system shall protect user accounts and personal information through secure authentication and access control.

Users shall only be able to access their own people, thoughts, memories, and associated information. Unauthorized users must not be able to access another user's private data.

9.3 Privacy

Privacy is a fundamental requirement of MIORA.

The system shall keep personal memories and information private by default. People, thoughts, memories, and contextual information shall not be publicly visible.

Any future sharing functionality shall require explicit action and permission from the user.

9.4 Usability

MIORA shall prioritize simplicity and ease of use.

The core action of recording a thought should require minimal interaction:

Select Person → Record Thought

Adding memories, conversations, places, or feelings shall remain optional and must not create unnecessary barriers to recording a thought.

9.5 Responsiveness

MIORA shall provide a responsive interface that functions across different screen sizes, including:

- Mobile devices
- Tablets
- Desktop browsers

The user experience shall prioritize mobile usability because moments of remembrance can occur at any time and location.

9.6 Reliability and Data Preservation

The system shall prioritize the reliable preservation of personal memories and thought records.

Accidental data loss should be minimized through confirmation mechanisms, archive functionality, and controlled deletion processes.

Archived people and their associated information shall remain recoverable unless the user explicitly completes the permanent deletion process.

9.7 Scalability

The initial version of MIORA shall be designed to support future growth in:

- Number of users
- Number of people per user
- Number of thought events
- Amount of contextual information

The system architecture should allow future expansion without requiring a complete redesign.

9.8 Future Non-Functional Enhancements

The following capabilities are outside the scope of the initial version but may be considered for future development:

Offline Support

Future versions may allow users to record thoughts without an internet connection and synchronize the information when connectivity becomes available.

Notification Support

Future versions may provide optional reminders or reflection notifications. These features should remain configurable and should not interfere with the natural, thought-driven nature of MIORA.

10. System Design

The system design of MIORA defines the conceptual structure, core entities, workflows, data flow, and high-level technical architecture of the proposed system. The design follows a people-centered approach in which users maintain meaningful people and record moments when those people come to mind.

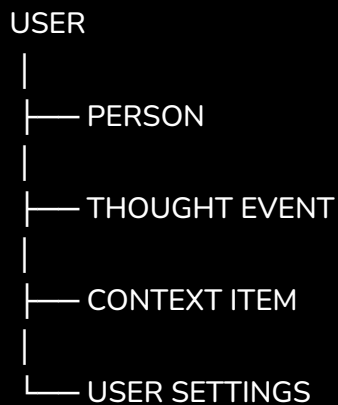
The system is designed to remain simple for the user while maintaining clear data relationships, ownership rules, and reliable handling of memories and deletion processes.

10.1 Core Domain Model

MIORA is built around five primary concepts:

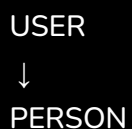
- User
- Person
- Thought Event
- Context Item
- User Settings

The conceptual structure is:



A user owns their people and associated information. Each person can have multiple thought events, and each thought event can contain multiple optional context items.

The core data chain is:



↓
THOUGHT EVENT
↓
CONTEXT ITEM

10.2 Core Entity Structure

User

The user represents the owner of a MIORA account. User authentication and identity management are handled through a managed authentication service.

The authenticated user identity is used to ensure that users can access only their own information.

Person

A Person represents someone meaningful to the user.

The primary information associated with a person includes:

- Person ID
- User ID
- Name
- Nickname (optional)
- Status
- Creation timestamp
- Update timestamp

A person is intentionally kept simple because MIORA is not designed as a contact management system.

Thought Event

A Thought Event represents a genuine moment when a person comes to the user's mind.

Each Thought Event contains:

- Thought ID
- User ID
- Person ID

- Occurred date and time
- Recorded date and time
- Creation timestamp
- Update timestamp

One valid Thought Event represents one remembrance.

The total remembrance count of a person can therefore be derived from the number of valid Thought Events associated with that person.

Context Item

Context Items represent optional information associated with a Thought Event.

Each Context Item contains:

- Context ID
- User ID
- Thought ID
- Context type
- Content
- Creation timestamp
- Update timestamp

The initial supported context types are:

- Memory
- Conversation
- Place
- Feeling

The flexible Context Item model allows multiple pieces of context to be associated with a Thought Event and allows future context types to be added without redesigning the core data structure.

User Settings

User Settings store configurable preferences associated with a user.

The initial setting includes:

- User ID
- Thought cooldown period

The default thought cooldown period is five minutes.

10.3 Data Ownership and Integrity

MIORA follows a consistent ownership model.

PERSON

└─ user_id

THOUGHT EVENT

└─ user_id

└─ person_id

CONTEXT ITEM

└─ user_id

└─ thought_id

USER SETTINGS

└─ user_id

The following integrity rules apply:

- 1. A person must belong to a valid user.
- 2. A Thought Event must belong to the same user as its associated Person.
- 3. A Context Item must belong to the same user as its associated Thought Event.
- 4. Users must not access or modify information belonging to another user.

These rules provide clear ownership and simplify authorization and user-level data access.

10.4 Record Thought Workflow

Recording a thought is the central interaction in MIORA.

The system prioritizes immediate thought capture.

The primary workflow is:

PEOPLE-FIRST HOME

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SELECT PERSON

↓

PERSON PROFILE



"I THOUGHT ABOUT THEM"



CHECK REPEAT / COOLDOWN



SAVE THOUGHT



THOUGHT RECORDED

For a normal thought, the Thought Event is saved immediately.

By default:

- Occurred time = Current time
- Recorded time = Current time

After recording, users may optionally:

- Add context
- Modify the occurred date and time
- Undo the thought

This approach ensures that recording a thought requires minimal interaction.

Repeated Thought and Cooldown Handling

MIORA allows multiple genuine thoughts about the same person, including multiple thoughts on the same day.

The default cooldown period is five minutes.

When a user attempts to record another thought within the configured cooldown period, the system asks for confirmation.

USER TAPS THOUGHT ACTION



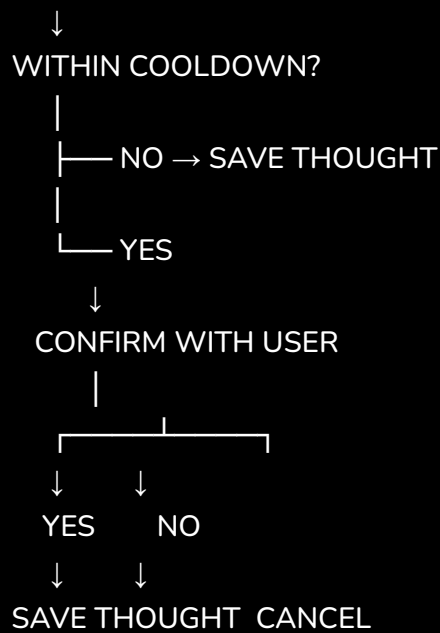
RECENT THOUGHT EXISTS?



| — NO → SAVE THOUGHT



| — YES



This prevents accidental repeated interactions while still allowing genuine multiple moments of remembrance.

Thought Deletion

Users can undo or delete an individual Thought Event.

Deletion requires confirmation.

When confirmed:

- The Thought Event is permanently deleted.
- All associated Context Items are permanently deleted.
- The person's remembrance count automatically reflects the remaining Thought Events.

10.5 Context Management Workflow

Context is optional and secondary to the Thought Event.

A Thought Event can exist without context, but a Context Item cannot exist without a Thought Event.

THOUGHT EVENT

↓
0 TO MANY
↓

CONTEXT ITEMS

Users can add multiple Context Items of different types.

The workflow is:

SELECT THOUGHT EVENT



ADD CONTEXT



SELECT TYPE



ENTER CONTENT



SAVE

Users can add multiple:

- Memories
- Conversations
- Places
- Feelings

Each Context Item supports free-text input.

Users can also edit or permanently delete individual Context Items without deleting the associated Thought Event.

10.6 Person Lifecycle Workflow

MIORA uses a controlled lifecycle for people to protect emotionally meaningful information from impulsive deletion.

The lifecycle is:

ACTIVE

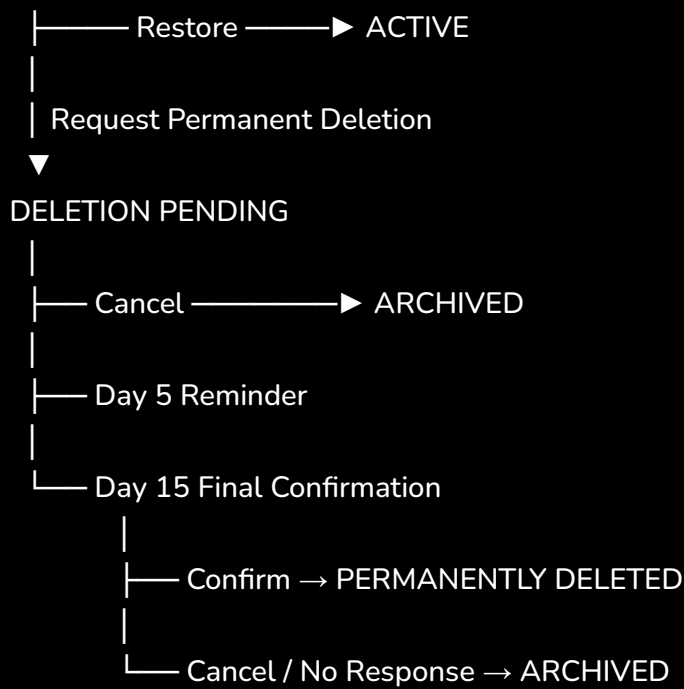


| Archive



ARCHIVED





Active

An active person:

- Appears in the main people list.
- Can receive new Thought Events.
- Has an accessible profile and timeline.

Archived

When archived:

- The person is removed from the active people list.
- No information is deleted.
- Existing Thought Events and Context Items remain preserved.
- The person appears in the Archived People section.
- The person can still appear in search results with an Archived status.
- New Thought Events cannot be recorded until the person is restored.

Restore

An archived person can be restored at any time.

Restoring returns the person to the Active state without losing any previous information.

Deletion Pending

A permanent deletion request can only begin when a person is archived.

The person enters a Deletion Pending state.

During this period:

- Information remains preserved.
- The user can cancel the request at any time.
- A reminder is provided after five days.
- A final confirmation is requested after fifteen days.

Permanent deletion requires explicit confirmation.

If the request is cancelled or the user does not provide final confirmation, the person returns to the Archived state.

Permanent Person Deletion

When permanent deletion is explicitly confirmed:

PERSON



THOUGHT EVENTS



CONTEXT ITEMS

The Person, all associated Thought Events, and all associated Context Items are permanently deleted.

10.7 High-Level System Architecture

MIORA follows a simple layered architecture.

USER



FRONTEND / CLIENT



AUTHENTICATED API REQUEST



BACKEND / API

↓
DATABASE

Frontend

The frontend is responsible for:

- People-first interface
- Person profiles
- Thought recording
- Context management
- Timelines
- Search
- Settings
- User interaction and feedback

The frontend does not enforce critical security or business rules independently.

Backend

The backend acts as the main business logic layer.

It is responsible for:

- Authorization
- Ownership validation
- Thought management
- Cooldown handling
- Context management
- Person lifecycle management
- Deletion workflows
- Input validation

Database

The database provides persistent storage for:

- People
- Thought Events
- Context Items
- User Settings

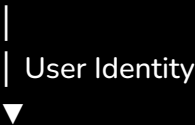
The database maintains relationships and data integrity between these entities.

10.8 Data Storage Architecture

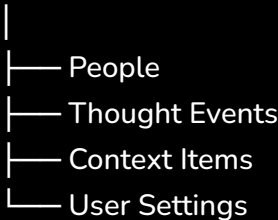
MIORA uses a relational data architecture because the system contains strongly connected entities and clear ownership relationships.

The conceptual structure is:

MANAGED AUTHENTICATION



POSTGRESQL DATABASE



A relational database supports:

- Clear relationships
- Foreign key constraints
- Data integrity
- Timeline queries
- Search
- Archive workflows
- Cascading deletion
- Long-term scalability

PostgreSQL is selected as the primary database technology.

10.9 Technology Architecture

The initial MIORA technology stack is:

Layer	Technology
-------	------------

Frontend	React
Frontend Language	TypeScript
Styling	Tailwind CSS
Backend	FastAPI
Backend Language	Python
Authentication	Managed Authentication
Database	PostgreSQL

The architecture follows the principle:

Simple frontend, intelligent backend, and reliable database.

The initial version avoids unnecessary architectural complexity such as microservices or multiple databases.

10.10 System-Level Data Flow

The general system data flow is:



Add Person

USER



ADD PERSON



FRONTEND



BACKEND



VALIDATE USER + INPUT



CREATE PERSON



DATABASE



UPDATE PEOPLE LIST

Record Thought

USER



SELECT PERSON



"I THOUGHT ABOUT THEM"



FRONTEND



BACKEND



VALIDATE USER



VALIDATE PERSON OWNERSHIP



CHECK PERSON STATUS



CHECK COOLDOWN



CREATE THOUGHT EVENT



DATABASE



RETURN SUCCESS



THOUGHT RECORDED

Add Context

USER



SELECT THOUGHT



ADD CONTEXT



SELECT TYPE + ENTER CONTENT



BACKEND



VALIDATE OWNERSHIP



CREATE CONTEXT ITEM



DATABASE



TIMELINE UPDATED

Archive and Restore

ARCHIVE



VALIDATE OWNERSHIP



STATUS → ARCHIVED

RESTORE



VALIDATE OWNERSHIP



STATUS → ACTIVE

No associated memories or thoughts are deleted during archive or restore operations.

System Design Summary

The MIORA system is designed around a simple and people-centered structure:

USER

↓

PERSON

↓

THOUGHT EVENT

↓

CONTEXT ITEM

The system prioritizes immediate thought capture, optional contextual reflection, clear data ownership, long-term preservation, and controlled deletion.

The architecture combines a React and TypeScript frontend styled with Tailwind CSS, a FastAPI backend responsible for business logic, managed authentication, and a PostgreSQL relational database.

11. Detailed Database Design

This section defines the detailed database structure of MIOIRA. The database is designed around the core relationship between users, meaningful people, remembrance events, and optional contextual reflections.

The design uses PostgreSQL as the relational database and follows principles of clear ownership, data integrity, controlled deletion, and efficient querying.

The four core application tables are:

- 1. `people`
- 2. `thought_events`
- 3. `context_items`
- 4. `user_settings`

Authentication user information is managed separately through the selected managed authentication system.

11.1 People Table

The `people` table represents meaningful people added by a user.

Structure

Column	Type	Required	Description
<code>person_id</code>	UUID	Yes	Unique identifier for the person
<code>user_id</code>	UUID	Yes	Authenticated owner of the person
<code>name</code>	VARCHAR	Yes	Person's primary name
<code>nickname</code>	VARCHAR	No	Optional personal nickname
<code>status</code>	VARCHAR(30)	Yes	Current lifecycle status
<code>deletion_requested_at</code>	TIMESTAMP	No	Time at which permanent deletion was requested
<code>created_at</code>	TIMESTAMP	Yes	Time when the person was created

updated_at	TIMESTAMP	Yes	Last modification time
------------	-----------	-----	------------------------

Person Status

The allowed lifecycle values are:

ACTIVE

ARCHIVED

DELETION_PENDING

A PostgreSQL **CHECK** constraint is used to validate the allowed values.

A permanently deleted status is not stored because permanently deleted records no longer exist.

Deletion Timestamp Rules

ACTIVE

→ deletion_requested_at = NULL

ARCHIVED

→ deletion_requested_at = NULL

DELETION_PENDING

→ deletion_requested_at = deletion request timestamp

Deletion Cancelled

→ status = ARCHIVED

→ deletion_requested_at = NULL

11.2 Thought Events Table

The **thought_events** table represents individual moments when a person comes to the user's mind.

Each valid Thought Event represents one remembrance.

Structure

Column	Type	Required	Description
<code>thought_id</code>	UUID	Yes	Unique Thought Event identifier
<code>user_id</code>	UUID	Yes	Owner of the Thought Event
<code>person_id</code>	UUID	Yes	Associated person
<code>occurred_at</code>	TIMESTAMP	Yes	When the thought actually occurred
<code>recorded_at</code>	TIMESTAMP	Yes	When the thought was recorded in MIORA
<code>updated_at</code>	TIMESTAMP	Yes	Last modification time

Timestamp Behaviour

MIORA supports recording a thought after it occurred.

For example:

Thought occurred:
Yesterday at 8:00 PM

Thought recorded:
Today at 10:00 AM

Therefore:

`occurred_at` ≤ `recorded_at`

The remembrance count of a person can be derived from the number of associated Thought Events.

11.3 Context Items Table

The `context_items` table stores optional contextual information associated with a Thought Event.

A Thought Event can exist without context, but a Context Item cannot exist without an associated Thought Event.

Structure

Column	Type	Required	Description
context_id	UUID	Yes	Unique Context Item identifier
user_id	UUID	Yes	Owner of the context
thought_id	UUID	Yes	Associated Thought Event
type	VARCHAR(30)	Yes	Type of context
content	TEXT	Yes	User-entered content
created_at	TIMESTAMP	Yes	Context creation time
updated_at	TIMESTAMP	Yes	Last modification time

Supported Context Types

The initial version supports:

- MEMORY
- CONVERSATION
- PLACE
- FEELING

The context type is validated using a **CHECK** constraint.

Multiple Context Items of the same type are allowed.

For example, one Thought Event may contain multiple feelings, memories, or places.

11.4 User Settings Table

The `user_settings` table stores configurable user preferences.

Structure

Column	Type	Required	Description
<code>settings_id</code>	UUID	Yes	Unique settings identifier
<code>user_id</code>	UUID	Yes	Owner of the settings
<code>thought_cooldown_minutes</code>	INTEGER	Yes	Cooldown for repeated thought recordings
<code>created_at</code>	TIMESTAMP	Yes	Settings creation time
<code>updated_at</code>	TIMESTAMP	Yes	Last modification time

Default Setting

The default cooldown value is:

`thought_cooldown_minutes = 5`

Every new MIORA user automatically receives a User Settings record.

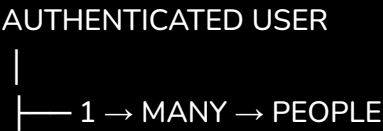
The `user_id` column is unique, ensuring that each user has only one settings record.

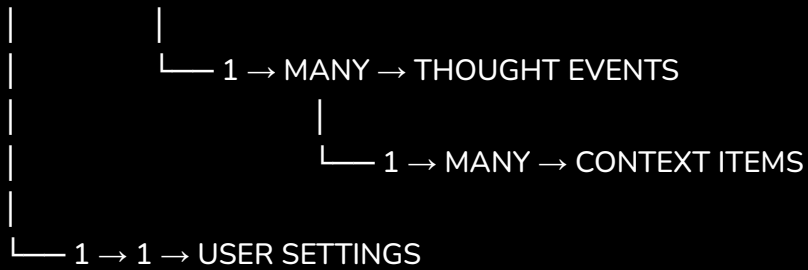
The cooldown value must satisfy:

`thought_cooldown_minutes >= 0`

11.5 Relationships, Keys and Referential Actions

The complete relationship structure is:





Primary Keys

people

└─ person_id

thought_events

└─ thought_id

context_items

└─ context_id

user_settings

└─ settings_id

Foreign Key Relationships

people.user_id

→ authenticated user

thought_events.user_id

→ authenticated user

thought_events.person_id

→ people.person_id

context_items.user_id

→ authenticated user

context_items.thought_id

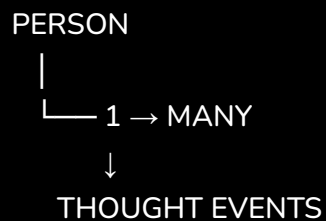
→ thought_events.thought_id

user_settings.user_id

→ authenticated user

Person to Thought Events

A person can have multiple Thought Events.



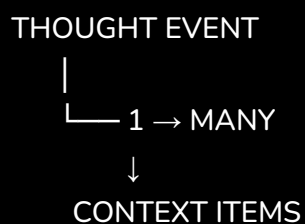
When a person is permanently deleted:



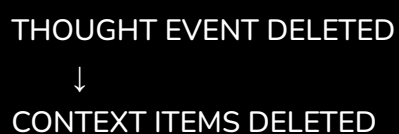
The relationship uses cascading deletion.

Thought Event to Context Items

A Thought Event can contain multiple Context Items.



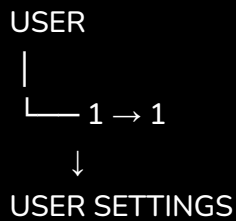
When a Thought Event is permanently deleted:



This relationship also uses cascading deletion.

User to User Settings

Each user has one settings record.



The uniqueness of `user_id` ensures that duplicate settings records cannot exist.

11.6 Database Constraints and Validation Rules

The database uses constraints to protect data integrity.

People

A person's name:

- Must not be NULL.
- Must not be empty.
- Must not contain only whitespace.

Conceptually:

`TRIM(name) <> ''`

The person status must be one of:

ACTIVE

ARCHIVED
DELETION_PENDING

The deletion timestamp must be consistent with the lifecycle status.

DELETION_PENDING
→ deletion_requested_at IS NOT NULL

ACTIVE / ARCHIVED
→ deletion_requested_at IS NULL

Thought Events

Every Thought Event must:

- Belong to a valid user.
- Belong to a valid person.
- Have an occurred timestamp.
- Have a recorded timestamp.

A thought may be recorded after it occurred:

$\text{occurred_at} \leq \text{recorded_at}$

Context Items

Every Context Item must:

- Belong to a valid user.
- Belong to a valid Thought Event.
- Have a valid type.
- Contain non-empty content.

Conceptually:

$\text{TRIM}(\text{content}) \neq ''$

Allowed types are:

MEMORY

CONVERSATION

PLACE

FEELING

User Settings

The settings table enforces:

UNIQUE(user_id)

and:

thought_cooldown_minutes >= 0

Ownership Consistency

MIORA stores **user_id** in multiple tables for direct ownership tracking.

The backend must ensure:

thought_events.user_id

=

people.user_id

and:

context_items.user_id

=

thought_events.user_id

Users must only access or modify records belonging to their authenticated identity.

11.7 Database Indexing Strategy

Indexes are selected based on MIOIRA's actual workflows.

The system avoids unnecessary indexing.

People

Fetch active people

```
WHERE user_id = ?  
AND status = 'ACTIVE'
```

Recommended index:

(user_id, status)

Search people

(user_id, name)

This provides a simple starting point for user-level name searches.

Thought Events

Person timeline

```
WHERE person_id = ?  
ORDER BY occurred_at DESC
```

Recommended index:

(person_id, occurred_at DESC)

Cooldown check

```
WHERE person_id = ?  
ORDER BY recorded_at DESC  
LIMIT 1
```

Recommended index:

(person_id, recorded_at DESC)

Context Items

Contexts are commonly retrieved using:

```
WHERE thought_id = ?
```

Recommended index:

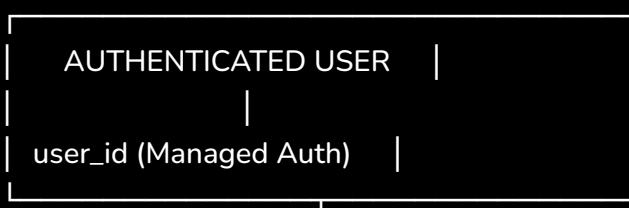
(thought_id)

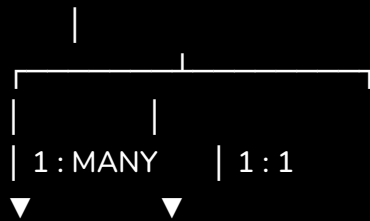
User Settings

The unique constraint on **user_id** automatically provides an index.

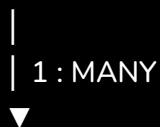
No additional index is required.

11.8 Complete Database Relationship Diagram

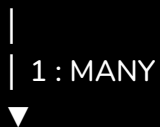




PEOPLE		USER_SETTINGS	
PK person_id		PK settings_id	
FK user_id		FK user_id UNIQUE	
name		cooldown_minutes	
nickname		created_at	
status		updated_at	
deletion_			
requested_at			
created_at			
updated_at			



THOUGHT_EVENTS	
PK thought_id	
FK user_id	
FK person_id	
occurred_at	
recorded_at	
updated_at	



CONTEXT_ITEMS	
PK context_id	
FK user_id	
FK thought_id	

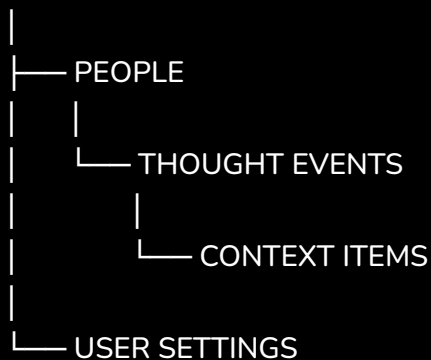
type	
content	
created_at	
updated_at	

Database Design Summary

The MIORA database is intentionally small and focused.

The complete structure is:

USER



This structure supports:

- A people-centered experience.
- Recording multiple genuine thoughts.
- Recording thoughts from the past.
- Optional memories, conversations, places, and feelings.
- Person archiving and restoration.
- Protected permanent deletion.
- Adjustable thought cooldowns.
- Clear user ownership.
- Long-term scalability.

The design provides a strong and simple foundation for the first version of MIORA while allowing additional features to be introduced in future versions.

12. API Design

This section defines the API design for MIORA. The API provides communication between the React frontend and the FastAPI backend.

The API follows a simple REST-based architecture and is designed around the core MIORA entities:

- People
- Thought Events
- Context Items
- User Settings

The API design prioritizes simplicity, clear resource ownership, consistent behaviour, and secure access to user data.

12.1 API Architecture

The API uses versioned endpoints with the following base path:

`/api/v1`

The high-level API structure is:

```
/api/v1
|
├── people
├── thoughts
├── contexts
└── settings
```

The general principle is:

Simple CRUD operations

↓

Standard REST endpoints

Business or lifecycle actions



Dedicated action endpoints

This keeps normal resource operations RESTful while allowing MIORA-specific actions to clearly represent their business behaviour.

12.2 Authentication and Authorization

All protected API requests require an authenticated access token.

The authentication flow is:

USER



Managed Authentication



Access Token



React Frontend



Authorization: Bearer <token>



FastAPI



Validate Token



Extract Authenticated user_id



Process Request

User Identity

The frontend must never send `user_id` manually in request bodies.

For example:


```
{  
  "name": "Deepthi",  
  "nickname": "Moni"  
}
```

The backend extracts the authenticated user identity from the access token and assigns ownership internally.

This applies to:

- People
 - Thought Events
 - Context Items
 - User Settings
 - Archive actions
 - Deletion actions
-

Resource Ownership

Every protected request follows:

REQUEST

↓

VALIDATE TOKEN

↓

EXTRACT USER ID

↓

FETCH RESOURCE

↓

VERIFY OWNERSHIP

↓

ALLOW / DENY

A user can only access resources belonging to their authenticated identity.

When a requested resource does not belong to the authenticated user, the API returns:

404 Not Found

This prevents the API from revealing the existence of another user's resource.

12.3 People APIs

The People APIs manage meaningful people added by a user.

Create Person

POST /api/v1/people

Request

```
{
  "name": "Deepthi",
  "nickname": "Moni"
}
```

The backend automatically assigns:

```
person_id
user_id
status = ACTIVE
deletion_requested_at = NULL
created_at
updated_at
```

Response

201 Created

```
{
  "person_id": "uuid",
  "name": "Deepthi",
```

```
"nickname": "Moni",  
"status": "ACTIVE",  
"thought_count": 0,  
"created_at": "2026-08-30T10:00:00Z"  
}
```

Get People

GET /api/v1/people

By default, this endpoint returns active people.

GET /people



ACTIVE PEOPLE

Additional lifecycle filtering can be performed using:

GET /people?status=ARCHIVED

GET /people?status=DELETION_PENDING

Ordering

People are ordered by their most recent Thought Event.

People with thoughts



Most recently remembered first



People with no thoughts

This supports MIORA's people-centered experience by prioritizing people currently appearing in the user's thoughts.

Response

```
{
  "people": [
    {
      "person_id": "uuid-1",
      "name": "Deepthi",
      "nickname": "Moni",
      "status": "ACTIVE",
      "thought_count": 42,
      "last_thought_at": "2026-08-29T18:30:00Z"
    }
  ]
}
```

The Thought Event count is calculated from `thought_events` and is not stored as a duplicate counter in the `people` table.

Get a Single Person

GET /api/v1/people/{person_id}

Response

```
{
  "person_id": "uuid",
  "name": "Deepthi",
  "nickname": "Moni",
  "status": "ACTIVE",
  "thought_count": 42,
  "last_thought_at": "2026-08-29T18:30:00Z",
  "created_at": "2026-05-10T10:00:00Z",
  "updated_at": "2026-08-29T18:30:00Z"
}
```

People with the `DELETION_PENDING` status remain viewable so that users can review their memories and cancel the deletion process.

Update Person

PATCH /api/v1/people/{person_id}

The user can update:

- Name
- Nickname

Example Request

```
{  
  "nickname": "Moniii"  
}
```

Only the supplied fields are updated.

Person Lifecycle APIs

MIORA uses dedicated action endpoints for lifecycle operations.

Archive Person

POST /api/v1/people/{person_id}/archive

Behaviour:

ACTIVE

↓

ARCHIVED

Restore Person

POST /api/v1/people/{person_id}/restore

Behaviour:

ARCHIVED

↓

ACTIVE

Request Permanent Deletion

A person must be archived before requesting permanent deletion.

POST /api/v1/people/{person_id}/request-deletion

Behaviour:

ARCHIVED

↓

DELETION_PENDING

deletion_requested_at

↓

Current Timestamp

Cancel Deletion Request

POST /api/v1/people/{person_id}/cancel-deletion

Behaviour:

DELETION_PENDING

↓

ARCHIVED

deletion_requested_at



NULL

Confirm Permanent Deletion

POST /api/v1/people/{person_id}/confirm-deletion

Permanent deletion is enforced only after the required waiting period.

current_time

≥

deletion_requested_at + 15 days

The backend enforces this rule.

After successful confirmation:

PERSON DELETED



THOUGHT EVENTS DELETED



CONTEXT ITEMS DELETED

The API returns:

204 No Content

The deletion workflow is:

ARCHIVED



Request Permanent Deletion



DELETION_PENDING



Day 5 → Reminder



Day 15 → Final Confirmation Available



User Explicitly Confirms



Permanent Deletion

12.4 Thought APIs

Thought APIs manage individual remembrance events.

Record a Thought

POST /api/v1/people/{person_id}/thoughts

For a current thought:

```
{}
```

The backend assigns:

`occurred_at = NOW`

`recorded_at = NOW`

For a past thought:

```
{  
  "occurred_at": "2026-08-29T20:00:00Z"  
}
```

The backend sets:

`occurred_at = user-provided time`

`recorded_at = current time`

A past thought can occur before the person was added to MIORA.

However:

$\text{occurred_at} \leq \text{recorded_at}$

must always be satisfied.

Cooldown Handling

MIORA uses an adjustable cooldown to prevent accidental repeated taps.

The default cooldown is five minutes.

POST Thought

↓

Check latest thought

↓

Within cooldown?

If no recent Thought Event exists:

201 Created

If a recent Thought Event exists:

```
{  
  "detail": "A recent thought was already recorded.",  
  "confirmation_required": true  
}
```

The frontend asks:

Did you remember them again?

If the user confirms, the request is repeated with:

```
{
```

```
"confirm_repeat": true
}
```

The backend then creates the Thought Event.

`confirm_repeat` bypasses only the cooldown confirmation and does not bypass other validation or authorization checks.

Get Person Thoughts

GET /api/v1/people/{person_id}/thoughts

Thought Events are returned as lightweight summaries.

They are ordered by:

occurred_at DESC

Response

```
{
  "thoughts": [
    {
      "thought_id": "uuid-1",
      "occurred_at": "2026-08-30T10:00:00Z",
      "recorded_at": "2026-08-30T10:00:00Z",
      "context_count": 2
    }
  ]
}
```

Context details are fetched separately when required.

Update Thought

PATCH /api/v1/thoughts/{thought_id}

The editable field is:

`occurred_at`

Example:

```
{  
  "occurred_at": "2026-08-28T18:30:00Z"  
}
```

`recorded_at` is permanent and cannot be changed.

Delete Thought

DELETE `/api/v1/thoughts/{thought_id}`

Deleting a Thought Event also deletes all associated Context Items.

DELETE THOUGHT



DELETE ASSOCIATED CONTEXT ITEMS

Response:

204 No Content

12.5 Context APIs

Context Items provide optional information associated with a Thought Event.

Supported types are:

MEMORY

CONVERSATION
PLACE
FEELING

Multiple Context Items of the same type are allowed.

Create Context Item

POST /api/v1/thoughts/{thought_id}/contexts

Request

```
{  
  "type": "MEMORY",  
  "content": "We spent the entire evening talking and laughing."  
}
```

Response

201 Created

```
{  
  "context_id": "uuid",  
  "thought_id": "uuid",  
  "type": "MEMORY",  
  "content": "We spent the entire evening talking and laughing.",  
  "created_at": "2026-08-30T10:00:00Z"  
}
```

Contexts are created separately after a Thought Event has been created.

The flow is:

Create Thought



Receive thought_id



Optionally create Context Items

Get Context Items

GET /api/v1/thoughts/{thought_id}/contexts

Contexts are returned in the order they were created:

ORDER BY created_at ASC

Update Context Item

PATCH /api/v1/contexts/{context_id}

The user can update:

- `type`
- `content`

Example:

```
{
  "type": "FEELING",
  "content": "Happy and nostalgic"
}
```

Delete Context Item

DELETE /api/v1/contexts/{context_id}

Only the selected Context Item is deleted.

The associated Thought Event remains.

Response:

204 No Content

12.6 User Settings APIs

The first version of MIORA contains one primary configurable setting:

thought_cooldown_minutes

Get Settings

GET /api/v1/settings

Response

```
{  
  "thought_cooldown_minutes": 5  
}
```

Update Settings

PATCH /api/v1/settings

Request

```
{  
  "thought_cooldown_minutes": 10  
}
```

The backend validates:

thought_cooldown_minutes $\geq$ 0

No unnecessary settings are introduced in the first version.

12.7 Common API Response and Error Handling

MIORA uses simple and consistent API responses.

Success Responses

Status Code	Meaning
200 OK	Request completed successfully
201 Created	A new resource was created
204 No Content	Resource was successfully deleted

Successful requests return the relevant resource data directly.

Error Responses

Errors use a simple structure:

```
{
  "detail": "Person not found."
}
```

Common Error Status Codes

Status Code	Usage
400 Bad Request	Invalid request or invalid resource state

401 Unauthorized	Missing or invalid authentication
404 Not Found	Resource does not exist or does not belong to the user
409 Conflict	Cooldown conflict or lifecycle conflict
422 Validation Error	Request data fails validation
500 Internal Server Error	Unexpected server-side error

Example cooldown response:

```
{
  "detail": "A recent thought was already recorded.",
  "confirmation_required": true
}
```

Example deletion waiting response:

```
{
  "detail": "Permanent deletion confirmation is not available yet."
}
```

FastAPI's standard validation responses are used for request validation errors.

API Design Summary

The final API structure is:

```
/api/v1
|
|— people
|   |— Create
|   |— List
|   |— Details
|   |— Update
```




The API design provides a simple and secure interface between the MIORA frontend and backend while supporting the complete first-version functionality of the system.

